

My approach is an AI/ML-based object detection method designed to track the ultrasound probe in each frame of a video. Since no publicly available dataset existed for this specific task, I manually annotated images to create a dataset from scratch. To increase dataset diversity and improve model generalization, I applied data augmentation techniques such as flipping, rotation, brightness adjustment, and motion blur. Given the relatively small dataset size, I opted for YOLOv8-nano, a lightweight object detection model optimized for efficiency and real-time inference.

To achieve high detection accuracy, I fine-tuned YOLOv8-nano, carefully selecting hyperparameters such as the number of training epochs and implementing early stopping to prevent overfitting. Overfitting was an initial challenge, which I mitigated by closely monitoring the validation loss and employing regularization techniques. The final model achieved a **mAP50 of 0.9887, precision of 0.97794, and recall of 0.96227**, indicating strong detection performance. While further improvements could be achieved by expanding the dataset and incorporating more varied scenarios, this approach provides an effective solution for real-time ultrasound probe detection using deep learning.